

TEMA (B) 1K13

$$1) a) \lim_{x \rightarrow 1} \frac{\sqrt{2x+2} - 2}{x-1} = \lim_{x \rightarrow 1} \frac{(\sqrt{2x+2} - 2)(\sqrt{2x+2} + 2)}{(x-1)(\sqrt{2x+2} + 2)}:$$

$$= \lim_{x \rightarrow 1} \frac{2x+2-4}{(x-1)(\sqrt{2x+2} + 2)} = \lim_{x \rightarrow 1} \frac{2x-2}{(x-1)(\sqrt{2x+2} + 2)} = \lim_{x \rightarrow 1} \frac{2(x-1)}{(x-1)(\sqrt{2x+2} + 2)} = \frac{2}{2+2} = \boxed{\frac{1}{2}}$$

$$b) \lim_{x \rightarrow 0} (1+2x)^{\frac{3}{x}}$$

$$t = 2x$$

$$x = \frac{t}{2}$$

$$\lim_{t \rightarrow 0} (1+t)^{\frac{3}{\frac{t}{2}}} = \lim_{t \rightarrow 0} (1+t)^{\frac{6}{t}} = \lim_{t \rightarrow 0} \left[ (1+t)^{\frac{1}{t}} \right]^6 =$$

$$= \left[ \lim_{t \rightarrow 0} (1+t)^{\frac{1}{t}} \right]^6 = \boxed{e^6}$$

$$2) a) y = 5x^{-2} \cdot \ln(\cos x) \quad y = s \cdot t \quad y' = s' \cdot t + s \cdot t'$$

$$s = 5x^{-2} \quad s' = -10x^{-3}$$

$$t = \ln(\cos x) \quad t' = \frac{-\sin x}{\cos x}$$

$$\boxed{y' = (-10x^{-3}) \cdot \ln(\cos x) + (5x^{-2}) \cdot \left( -\frac{\sin x}{\cos x} \right)}$$

$$b) y = 32 \cos(5x+1)^{-x} + 2e \quad y = s + t \quad y' = s' + t'$$

→ multiplico el final por 32

$$s = (32) \cos(5x+1)^{-x} = 32 \cos u \rightarrow u = (5x+1)^{-x} \quad \ln u = \ln(5x+1)^{-x}$$

$$\ln u = (-x) \ln(5x+1) \rightarrow \frac{u'}{u} = (-1) [\ln(5x+1)] - x \left( \frac{5}{5x+1} \right) \rightarrow u' = \left[ \ln(5x+1) - \left( \frac{5x}{5x+1} \right) \right] (5x+1)^{-x}$$

$$s' = 32 u' \cdot \sin u \quad s' = (-32) \cdot \left[ -\ln(5x+1) - \left( \frac{5x}{5x+1} \right) \right] (5x+1)^{-x} \cdot \sin(5x+1)^{-x}$$

$$t = 2e \quad t' = 0$$

$$\boxed{y' = (-32) \left[ -\ln(5x+1) - \left( \frac{5x}{5x+1} \right) \right] (5x+1)^{-x} \cdot \sin(5x+1)^{-x} + 2e}$$

$$c) \quad y = \frac{\cos(\ln x)}{\sqrt{\tan 2x}} \quad y = \frac{s}{t} \quad y' = \frac{s' \cdot t - s \cdot t'}{t^2}$$

$$s = \cos(\ln x) \quad s' = -\frac{1}{x} \sin(\ln x)$$

$$t = (\tan 2x)^{\frac{1}{2}} \quad t' = \frac{1}{2} (\tan 2x)^{-\frac{1}{2}} \cdot \frac{2}{\cos^2 2x} = \frac{(\tan 2x)^{-\frac{1}{2}}}{\cos^2 2x}$$

$$y' = \frac{(-\frac{1}{x}) \sin(\ln x) \cdot (\tan 2x)^{\frac{1}{2}} - \cos(\ln x) \cdot \frac{(\tan 2x)^{-\frac{1}{2}}}{\cos^2 2x}}{\tan 2x}$$